564-06-010 ISMRM ABSTRACT

2D U-Net Segmentation of Multi-target Deep Brain Stimulation Lead Trajectories on Postoperative Imaging

Digital Poster · 55 min View Poster | Segmentation for Neuro Applications · Session: 4:55–5:50 PM · Digital Posters Row E · 10 **Keywords:** SEGMENTATION DEEP BRAIN STIMULATION DEEP LEARNING SEGMENTATION SURGICAL PLANNING PRECISION NEUROMODULATION TARGETING

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Impact

No reliable open-source tools exist to segment deep brain stimulation lead trajectories. Leveraging deep learning to achieve fast yet reliable segmentation, our tool can be applied widely to inform surgical targeting strategies and to better understand variance in clinical outcomes.

Synopsis

Motivation: Response to deep brain stimulation (DBS) may depend on the entire lead trajectory, not just electrode contact locations. To understand this relationship, a robust lead segmentation tool is needed.

Goals: Develop an automated, open-source tool for accurate lead segmentation from postoperative CT and MRI.

Approach: A 2D U-Net model was trained and validated on 104 postoperative CT volumes from patients with DBS implants. Manual segmentations served as ground truths.

Results: On the test set (n=20), average slice- and volume-based Dice scores were 0.78±0.001 and 0.80±0.067 with minimal false negatives and positives. The model showed promising results for segmenting leads from postoperative MRI images.

Introduction

Deep brain stimulation (DBS) can be an effective therapy for managing symptoms associated with movement disorders and neuropsychiatric conditions. Despite benefiting many patients, there is still significant unexplained outcome variance. Lead placement has been shown to influence outcomes;[1–5] however, conventional measures of lead location only capture the position of electrode contacts at the lead tip, overlooking potential effects arising from the full implant trajectory (**Fig 1A**).

Lead trajectories (from the cranial entry point to the deep target) are dictated by the surgical strategy[6,7] and thus target selection, anatomy,[8] image quality, and surgical technology. As such, lead trajectories vary between and even within patients[9] (**Fig 1B**).

A 2013 trial reported worsened cognitive performance when DBS leads targeting the subthalamic nucleus (STN) also intersected the caudate.[10] A 2016 trial later showed persistent verbal fluency impairments after surgery, regardless of whether stimulation was initiated immediately, or after a 3-month delay, suggesting lead trajectory involvement through surgical injury.[11] Beyond these studies, lead trajectory effects remain largely unexplored. This gap (compared with extensive research on lead location enabled by open-source tissue activation modeling tools[12]) may reflect the historical lack of methods to map lead trajectories.

Two threshold-based segmentation approaches have been developed in small cohorts.[13,14] One deep learning approach developed a model using ground truth data automatically segmented from postoperative CTs (35 DBS and 45 sEEG cases), and achieved a Dice score of 0.92 with cross validation[15]. Building on this promising work, there remains a need for a robust, DBS-specific model that incorporates more patient data with manually annotated ground truths, and, crucially, that is open-source. From this, CT-to-MRI transfer learning can be used to develop an MRI-based tool that segments more challenging, artifact-prone leads on postoperative MRI (**Fig 1C**), providing a solution for patients without CT imaging.

Methods

Dataset Assembly: Postoperative CT volumes were collected for 124 patients with movement disorders and DBS targeting the basal ganglia, thalamus, or cerebellum. Each scan contained up to three leads, which appeared as small, high-contrast structures relative to surrounding brain parenchyma (**Fig 1C**). Ground truth annotations were generated in ITK-SNAP through meticulous, multi-rater manual segmentation, producing 3D binary masks delineating each patients' lead boundaries.

Data Preprocessing: Each 3D CT volume was processed slice-wise (with z-score normalization) to generate 2D training samples (512×512 pixel) and leverage the computational efficiency of 2D processing while maintaining the ability to reconstruct the full 3D lead masks through sequential slice aggregation.

Model Architecture: A 2D U-Net convolutional neural network was implemented using PyTorch. The U-Net architecture, originally designed for biomedical image segmentation,[16] features a contracting encoder path for context capture and an expanding decoder path for precise localization, connected through skip connections that preserve spatial information across scales. The model performs binary segmentation on individual CT slices, classifying each pixel as either DBS lead or background tissue. Sequential slice predictions are subsequently reconstructed to generate complete 3D binary masks representing lead volumes.

Training Protocol: The U-Net model was trained on a NVIDIA A100 high-performance GPU cluster, using a combination of Weighted Dice Loss (30%), Focal Loss (30%), and Binary Cross Entropy (40%) with a 70:15:15 train–validation–test split (Fig 2). Model optimization was achieved using the Adam optimizer with a learning rate of 1e-4 trained for 50 epochs with a batch size of 2.

Performance Measures: Prediction probability maps for each CT slice generated by the model were thresholded to create binary segmentation masks, with the threshold selected based on peak performance measures, relative to the ground truth masks. Model performance was then evaluated using the Dice score and confusion matrices.

Model testing on MRI: To advance toward developing an MR-based segmentation tool, the final CT-based model was applied to inverted, skull-stripped, postoperative T1-weighted images, and its performance was evaluated qualitatively.

Results

A threshold of 0.997 was shown to be optimal based on the F1 score (Fig 3), with (slice-wise) 31.0% True Positives, 0.2% False Positives, 68.6% True Negatives and 0.1% False Negatives (Fig 4). The average slice-wise Dice score for True Positives was 74.7% (Fig 4). Examples of accurate and inaccurate predictions are shown in (Fig 5). Poor predictions were observed in areas of suboptimal skull stripping. Promisingly, the CT-based model achieved good qualitative performance when applied to postoperative MRI images (Fig 5).

Discussion

Our model achieved robust lead segmentation with slice- and volume-based true positive Dice scores of 0.78±0.001 and 0.80±0.067, demonstrating good accuracy. To further enhance performance, we will expand the dataset, apply artificial data augmentation, and advanced class imbalance solutions. CT-to-MRI transfer learning will enable robust MRI-based segmentation.

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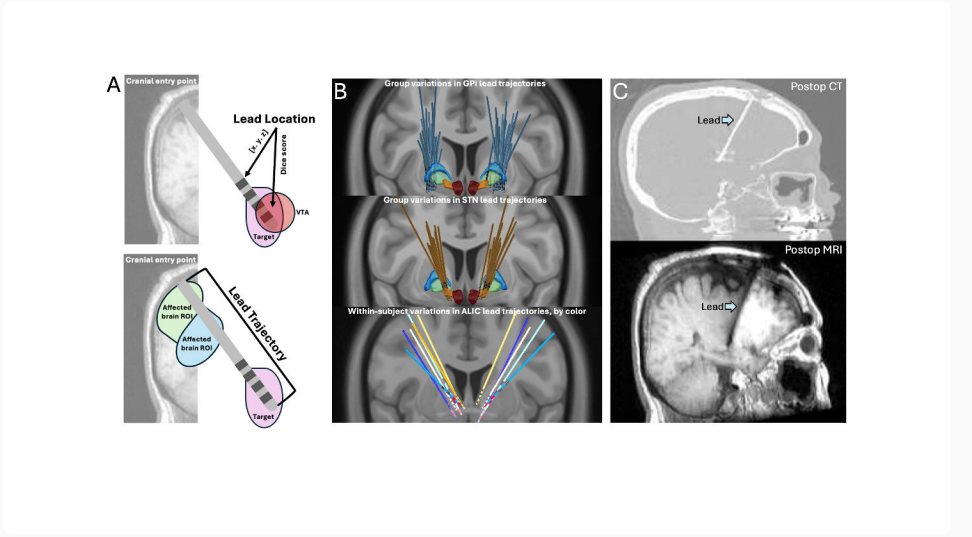
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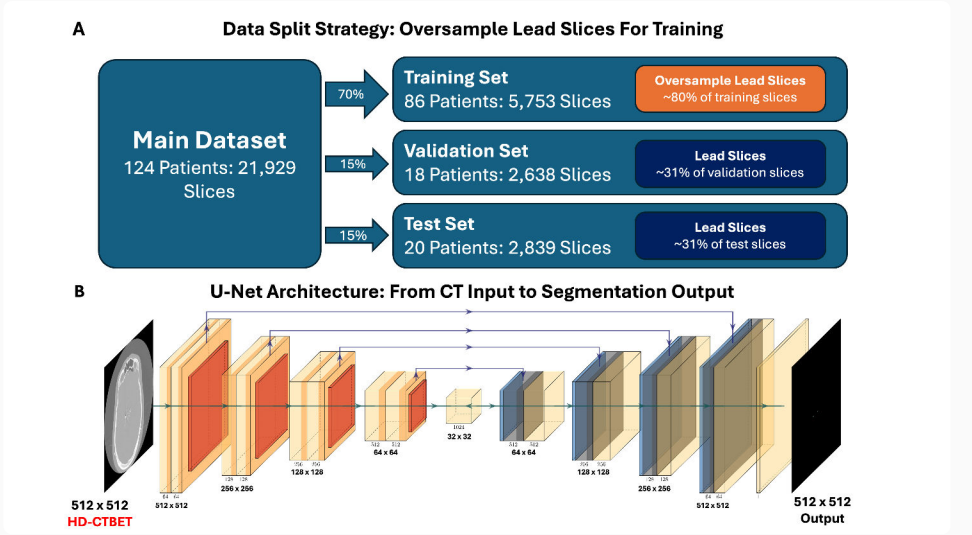
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Figures and Tables



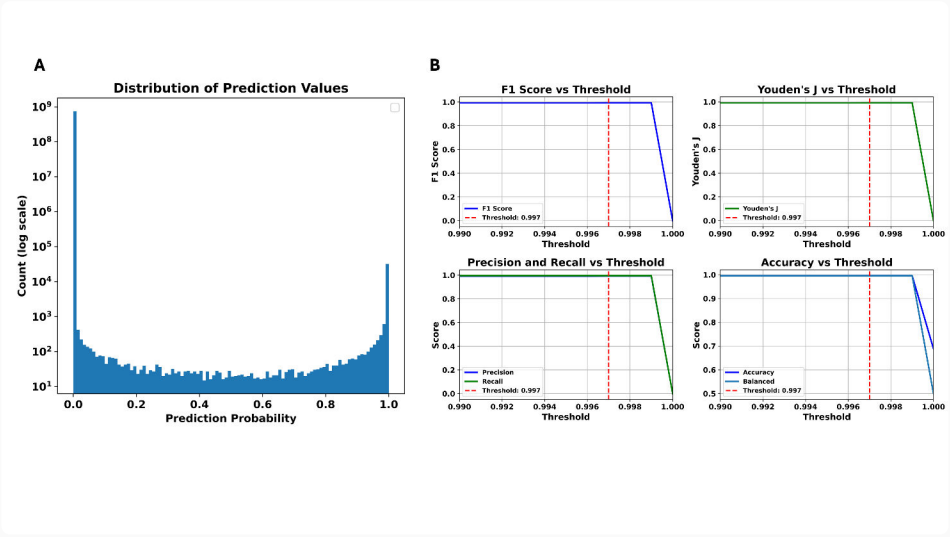
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Figure 1: *Schematic of DBS lead location versus lead trajectory (**A**). Examples of variations in lead trajectory across and within patients using (semi-)automated point-based reconstruction, not voxel-wise segmentation (**B**). Comparison of CT versus MRI lead segmentation problem (**C**).*



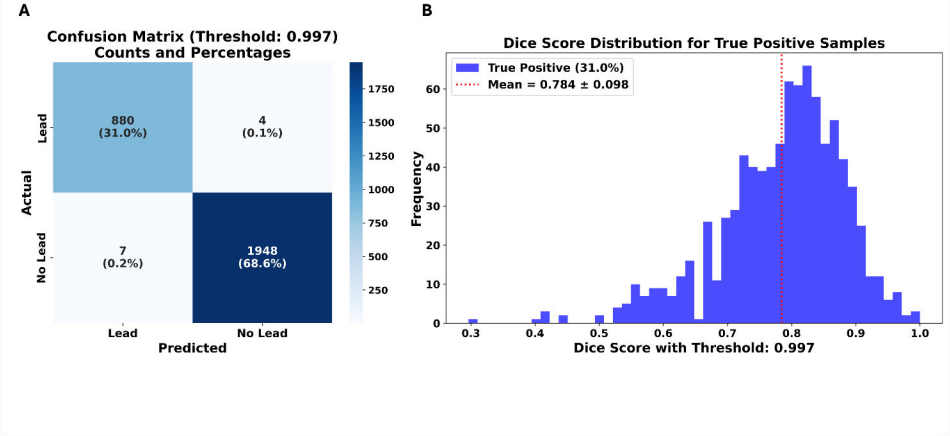
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Figure 2: *(**A**) CT images (21,929 slices) from all 124 patients were split into training (~70%), validation (~15%), and test (~15%) sets after skull stripping. Lead slices were oversampled in training set (~80%) to address the class imbalance. (**B**) The 2D U-Net architecture from the input CT image to predicted segmentation mask.*



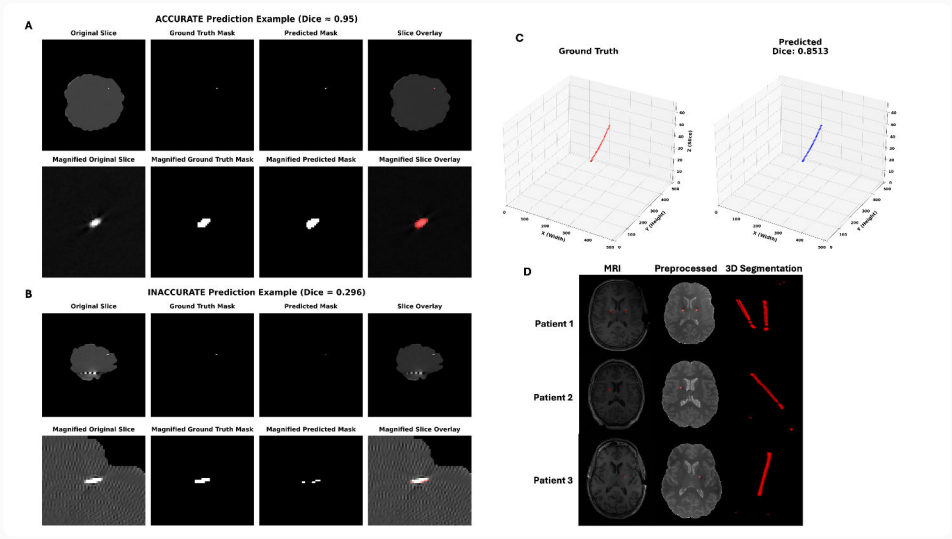
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Figure 3: *U-Net model prediction threshold analysis (**A**) Distribution of prediction probabilities across all 2,839 test slices. (**B**) Threshold optimization analysis showing F1 score, Youden's J, Precision/Recall, and Accuracy scores across span of threshold values. Threshold of 0.997 was shown to be most optimal based on F1 score.*



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Figure 4: *U-Net model performance with threshold of 0.997 (**A**) Confusion matrix demonstrating counts and percentages of True Positive, True Negative, False Positive, False Negative predictions on 2,839 test slices (**B**) True Positive Dice Score distribution.*



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Figure 5: *Representative examples of model performance, showing accurate (**A**) and inaccurate (**B**) 2D segmentation, and a 3D volume-based example (**C**). Performance of the CT-based model on MRI is also shown qualitatively (**D**).*